

Project Submission

# PriceSense

A SQL-Driven Pricing Intelligence Framework · Ai Palette × E-Cell IIT Guwahati 2026

TOTAL TRANSACTIONS

47,722

TOTAL UNITS SOLD

190,608

GROSS REVENUE

₹88.9L

AVERAGE SELLING PRICE

₹57

## THE PRICING BLIND SPOT

A D2C nutrition brand launching high-protein snacks has no data-backed price point. Higher prices kill demand; lower prices erode margin. Price sensitivity varies across fitness, budget, and premium personas and across geographies. This project constructs every pricing insight from first principles using raw SQL on transactional data.



# Methodology & SQL Framework

## TWO-PHASE ANALYTICAL APPROACH

### Phase 1 Sensitivity Framework

Price buckets built using `CAST(price/50 AS INTEGER)*50` floor math. `LAG()` window function computes inter-bucket % demand drop. Elasticity proxy is derived from first principles using actual transactional behavior, rather than clean assumptions.

$$\text{Elasticity Proxy} = \frac{\% \Delta Q}{\% \Delta P}$$

### Phase 2 Contextual Optimisation

Trend claims parsed with LIKE patterns. Revenue yields are calculated directly as `Revenue = Price × Quantity` aggregated across consumer dimensions (state, city\_tier, and occasion).

Independent `RANK()` `OVER()` partitions isolate the volume-maximizing price points versus revenue-maximizing thresholds.

## DATA QUALITY & FILTERING LOGIC

Assumption: bucket width of ₹50 balances granularity with statistical significance given the dataset's ~50k transactions. Negative-price rows (data entry errors) excluded via `WHERE price > 0 AND quantity > 0`.



# Overall Price Thresholds

## DEMAND BY ₹50 PRICE BUCKET

|          |               |          |
|----------|---------------|----------|
| ₹0-50    | 158,106 units | Baseline |
| ₹50-100  | 20,828        | -86.8%   |
| ₹100-150 | 8,298         | -60.2%   |
| ₹150-200 | 2,428         | -70.7%   |
| ₹200-250 | 298           | -87.7%   |

### THRESHOLD 1 — CLIFF

**₹50**

Demand falls 86.8% upon crossing ₹50.

### THRESHOLD 2 — FLOOR

**₹200**

Only 298 units sold above ₹200 (mass ceiling).

### ELASTICITY (₹0 → ₹50)

**-0.30**

Inelastic demand relative to higher tiers.

### ELASTICITY (₹150 → ₹200)

**-2.17**

Highly elastic regime; strong, clean drop.

## KEY FINDING

The critical demand cliff is at ₹50. A product priced above ₹50 loses ~87% of the volume base it would have at sub-₹50. The secondary cliff at ₹150 signals the premium ceiling for the mass-market segment.



# Price Sensitivity by Persona

## DEMAND DROP AT ₹50 THRESHOLD — BY PERSONA

### Budget

Units @ ₹0-50: 36,690

Units @ ₹50-100: 5,336

Drop at ₹50: -85.5%

Units @ ₹150-200: 390

Most sensitive above ₹150

### Casual

Units @ ₹0-50: 36,280

Units @ ₹50-100: 4,608

Drop at ₹50: -87.3%

Units @ ₹150-200: 646

Sharpest ₹50 cliff

### Fitness

Units @ ₹0-50: 38,015

Units @ ₹50-100: 6,033

Drop at ₹50: -84.1%

Units @ ₹150-200: 476

Best retention above ₹50

### Premium

Units @ ₹0-50: 38,115

Units @ ₹50-100: 3,732

Drop at ₹50: -90.2%

Units @ ₹150-200: 651

High ₹150+ retention

## KEY INSIGHTS

Fitness persona is the most price-tolerant above ₹50 (-84.1% drop vs -90.2% for premium). The premium persona, despite its label, shows the steepest drop at ₹50 — but sustains the highest absolute volume above ₹150 (651 units), suggesting tolerance for very high price points but resistance to the ₹50-100 mid-zone.



# Pricing Power of Claims

## AVERAGE REALISED PRICE BY CLAIM TIER

| CLAIM TIER                 | TRANSACTIONS | UNITS SOLD | AVG PRICE (₹) | GROSS REVENUE (₹) |
|----------------------------|--------------|------------|---------------|-------------------|
| Clean-Label Only           | 8,567        | 33,328     | ₹76.97        | 20,50,115         |
| High-Protein Only          | 7,618        | 35,077     | ₹61.98        | 16,81,416         |
| Baseline (No Claim)        | 29,582       | 1,15,525   | ₹51.13        | 49,45,894         |
| High-Protein + Clean-Label | 1,955        | 6,678      | ₹39.07        | 2,16,185          |

### FINDING

Clean-label products command an avg price of ₹76.97 — a **+50.5% PREMIUM** over baseline (₹51.13). High-protein adds a **+21.2% PREMIUM**. HP+CL combo SKUs have the lowest average price (₹39.07), suggesting they are budget combo bundles.

### RECOMMENDATION

Launch clean-label SKUs at ₹70-80 and high-protein SKUs at ₹60-70. Do not bundle both claims on a low-cost SKU — it dilutes premium perception.



# Pricing Power by Geography

## TOP REVENUE MARKETS (STATE × CITY TIER)

| STATE      | CITY TIER | UNITS SOLD | AVG PRICE (₹) | REVENUE (₹) | SIGNAL            |
|------------|-----------|------------|---------------|-------------|-------------------|
| Florida    | Tier 1    | 8,534      | ₹64.97        | 3,92,735    | Top revenue       |
| California | Tier 1    | 8,882      | ₹57.71        | 3,73,281    | High volume       |
| Colorado   | Tier 1    | 7,916      | ₹52.96        | 3,71,255    | -                 |
| Washington | Tier 2    | 4,695      | ₹86.18        | 3,60,188    | Highest avg price |
| Illinois   | Tier 2    | 6,045      | ₹75.00        | 3,61,981    | Strong Tier 2     |
| NY         | Tier 1    | 6,681      | ₹70.86        | 3,55,926    | -                 |

### FINDING

Tier 2 cities (Washington ₹86, Illinois ₹75) sustain *higher* average selling prices than most Tier 1 markets — suggesting less price competition and aspirational spending in smaller metros.

### RECOMMENDATION

Apply a Tier 2 price premium of ₹15-20 over Tier 1 MRP in Washington and Illinois. Price Florida and California Tier 1 competitively at ₹60-65 to capture raw volume.



# Occasion-Based Price Yield

## REVENUE YIELD PER UNIT BY OCCASION

|             |        |           |
|-------------|--------|-----------|
| Festive     | ₹52.86 | Premium   |
| Gym         | ₹48.56 | Mid-tier  |
| Marathon    | ₹47.96 | Mid-tier  |
| Road Trip   | ₹47.63 | Mid-tier  |
| Late Night  | ₹46.48 | High Vol  |
| On-the-go   | ₹46.04 | Mid-tier  |
| Daily Snack | ₹44.42 | Sensitive |
| Religious   | ₹42.39 | Most Sens |

### YIELD INSIGHTS

Festive occasion delivers the highest yield per unit (₹52.86) with 18,179 units. Religious fasting has the lowest yield (₹42.39) despite 28,056 units. Late-night is the highest-volume occasion (49,654 units) at ₹46.48/unit.

### STRATEGIC RECOMMENDATIONS

Launch premium festive gift packs at ₹75-90. Price daily-snack and religious-fasting SKUs at ₹35-45 to protect absolute consumer volume bases.



# Revenue vs. Volume Frontier

## PRICE BUCKET PERFORMANCE — VOLUME RANK VS REVENUE RANK

| PRICE BUCKET (₹) | AVG PRICE | TOTAL VOLUME | TOTAL REVENUE (₹) | VOL RANK | REV RANK | DIVERGENCE      |
|------------------|-----------|--------------|-------------------|----------|----------|-----------------|
| ₹0-50            | 17.18     | 1,58,106     | 28,99,124         | #1       | #1       | Aligned         |
| ₹50-100          | 67.41     | 20,828       | 14,20,153         | #2       | #2       | Aligned         |
| ₹100-150         | 126.45    | 8,298        | 10,56,215         | #3       | #3       | Aligned         |
| ₹150-200         | 167.73    | 2,428        | 4,11,711          | #4       | #4       | Aligned         |
| ₹200-250         | 212.06    | 298          | 60,009            | #5       | #11      | High Divergence |
| ₹14,450-14,500   | 14,463    | 15           | 2,16,945          | #24      | #5       | High Divergence |

### FINDING

For prices ₹0-200, revenue rank and volume rank are perfectly aligned. Above ₹200, high-value luxury SKUs (₹14,000+) generate outsized revenue with tiny volume — representing an entirely distinct business model.

### RECOMMENDATION

For mass-market launch, price at ₹0-150 to capture the aligned revenue-volume zone. Introduce a separate premium gifting SKU at ₹1,000-1,500 range where small volumes still generate high revenues.



# Launch Pricing Architecture

## RECOMMENDED LAUNCH PRICE ARCHITECTURE

| SEGMENT           | RECOMMENDED RANGE | CLAIM TARGET        | OCCASION TARGET                | CITY TIER       |
|-------------------|-------------------|---------------------|--------------------------------|-----------------|
| Budget            | ₹30-50            | Baseline / No claim | Daily snack, religious fasting | Tier 1 & 2      |
| Casual / Fitness  | ₹55-80            | High-protein        | Gym, marathon prep, on-the-go  | Tier 1          |
| Fitness (Premium) | ₹70-90            | Clean-label         | Gym, road trip                 | Tier 2 (WA, IL) |
| Premium / Gifting | ₹1,000-1,500      | Clean-label + HP    | Festive, gifting               | Tier 1 (FL, NY) |

### Volume Max

Price at ₹0-50. Captures 83% of all units. Low margin per unit — requires scaling.

### Revenue Max

₹50-150 is the sweet spot. Clean-label claim adds +50% average premium.

### Margin Max

Premium gifting SKUs at ₹1,000-1,500 yield outsized individual unit revenue.

### Next Steps

Run A/B price tests (₹49 vs ₹59). Activate clean-label claims in Tier 2 cities.



End of Presentation

# Questions?

SQL-Driven Pricing Intelligence Framework

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**Ai Palette x E-Cell**

IIT Guwahati 2026

**PriceSense Framework**

Raw SQL Insights